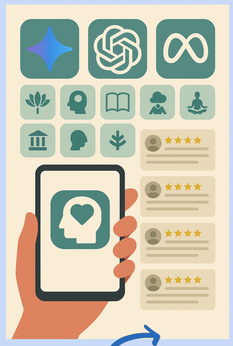
LEVERAGING LARGE LANGUAGE MODELS FOR REVIEW CLASSIFICATION AND RATING ESTIMATION OF MENTAL HEALTH APPLICATIONS

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INTRODUCTION



- Approximately **970,000,000 people** are affected by mental health conditions globally, mobile applications (**Apps**) provide scalable support to individuals who lack access to traditional care
- **10,000 Apps** are available on the market, individuals rely on **user reviews** to assess their quality & make a decision
- Applying **Large Language Models (LLMs)** to mental health text classification is relatively new and developing area of research
- Existing methods rely on supervised learning, deep learning, reinforcement learning, which heavily depend on **fine-tuning** and reward models.



PURPOSE

"Benchmarking Large Language Models (LLMs) for Review Classification "

With the long-term vision of building an **app recommendation system** from reviews, we first evaluate whether **self-supervised LLMs** can accurately classify **1–5 star ratings.** Focusing on mental health app reviews, we compare LLMs to **supervised ML models** and assess their ability to estimate overall app quality. We expect findings to generalize to other review platforms.

RESEARCH QUESTION (RQ)

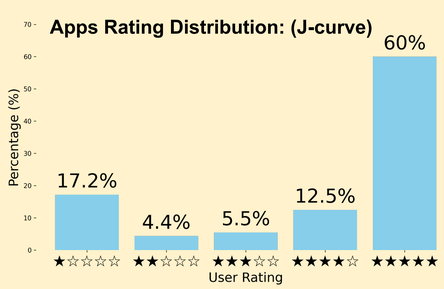
- RQ1: How do **LLMs** compare to **supervised models** in classifying mental health app reviews?
- RQ2: How do **LLM predictions** affect **app rankings**?
- RQ3: Do user ratings reflect true app **quality** or show **bias**?
- RQ4: What factors cause **rating discrepancies** in mental health apps?

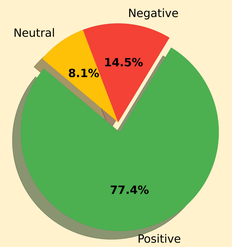
DATASET- MHARD

Dataset

- Mental Health App Review Dataset (**MHARD**)
- From Google Play Store, **collected** using SerpAPI
- Dimension: **200,973** rows x 8 features (date, user reviews, ratings, reviews, response, likes, etc.)
- App: **73** unique mental health apps with over 100,000 downloads
- Date Range: March 29, 2011 – July 11, 2023

App	Rating	Date	Review
Stoppanic	5	2/3/17	It really helps me, especially the relaxing audios really calm me down in this stressful world. Totally recommend this app.
Daylio	1	6/8/20	It's got good premise but the notifications don't work properly and the customer service were rubbish
Gratitude	5	11/25/18	This app is user friendly and easy to maintain. I love it!
Headspace	1	9/9/20	There is no point in me using this anymore if you guys are going to lock the SINGLE soundscape I use to destress and fall asleep!





METHODS

Data Collection & Preprocessing
Non-English texts (8.16%) Removed
PII Anonymized during text cleaning

Model Selection
Supervised Learning
Traditional ML Algorithms, Deep Learning
Self-Supervised LLM
OpenAI, Google, Meta

Evaluation
Precision, recall, F1-score, and accuracy were computed for **individual** reviews prediction, then **aggregated** at the app level.

Human Feedback
Recruit **crowdsource** workers to review cases where the GPT predict ratings **differed** from the original user ratings, we investigated using **thematic analysis**

ANALYSIS & RESULTS

Prompt: "You are an Expert APP RATER specializing in MENTAL HEALTH applications. Your task is to ACCURATELY CLASSIFY the review into one of the rating classes where 1 being the lowest and 5 being the highest. Only return a single number from Classes: [1, 2, 3, 4, 5] {individual user review}"

Feasible & Efficient
- **80%** accuracy from best supervised learning (BERT)
- **75%** accuracy from best LLMs (GPT-4 and LLaMA 3.3 70B)
- Suggests LLMs could be **high-fidelity annotators** for strong performance in multi-class review classification, without the need for extensive fine-tuning or reward modeling

Aggregated App Instance
- **GPT-4** outperforms supervised models with **significantly higher mean** in Recall (t = −6.66), Precision (t = −4.28), & F1-Score (t = −5.48), all p < .001 across app-level predictions
- GPT-4 identified **8 out of the top 10** apps (Gratitude, Moodpress, Intellect, Breath Ball, Healthy Minds, Dare, Mood Tracker, Daylio Journal), showing strong ranking recall

Embedding	Classifier	Precision	Recall	F1-Score	Accuracy
FasText	KNN	0.58	0.63	0.60	0.63
	SVM	0.63	0.73	0.66	0.73
	RF	0.73	0.69	0.61	0.69
	NN	0.67	0.59	0.62	0.59
	DT	0.56	0.55	0.56	0.55
MiniLM	KNN	0.65	0.72	0.67	0.72
	SVM	0.67	**0.74**	0.66	**0.74**
	RF	**0.75**	0.73	0.65	0.73
	NN	0.70	0.66	0.68	0.66
	DT	0.60	0.60	0.60	0.60
BERT-Large	KNN	0.59	0.66	0.62	0.66
	SVM	0.65	0.73	0.65	0.73
	RF	0.73	**0.74**	0.67	**0.74**
	NN	0.73	0.67	**0.70**	0.67
	DT	0.61	0.60	0.61	0.61
BERT (Base)		0.76	**0.80**	0.77	**0.80**
BERT (Large)		0.77	**0.80**	0.78	**0.80**
RoBERTa		**0.78**	**0.80**	**0.79**	**0.80**
GPT-3.5 Instruct		0.78	0.69	0.72	0.69
GPT-3.5 Turbo		0.78	0.67	0.71	0.67
GPT-4		**0.79**	**0.75**	**0.76**	**0.75**
Gemini 1.5 Flash		0.76	0.62	0.65	0.62
Gemini 1.5 Pro		**0.79**	0.74	**0.76**	0.74
Llama 3.1 (8B)		0.74	0.61	0.61	0.61
Llama 3.3 (70B)		0.77	**0.75**	0.75	**0.75**

Performance comparison across 15 traditional ML models, 3 deep learning models, and 7 LLMs. Bolded values indicate top scores per category per metric. Classifier abbreviations: KNN (K-Nearest Neighbors), RF (Random Forest), DT (Decision Tree), SVM (Support Vector Machine), NN (Neural Networks).

DISCUSSION

Original User: 4, GPT: 5
"Well, this app gives a new type of experience. I found this app to have a positive impact on my life."
Original User: 5, GPT: 3
"I like where this is taking me but I can't help but think that would these make any changes for me or me emotionally

Subjectivity:
- Users can **update reviews,** but original **ratings remain** unchanged on the Play Store.
- App users often select lower ratings when apps had **paid subscriptions**, even if they found them useful
- GPT likely interpret **3-star** reviews as **positive**
- Crowdsourcing via Amazon Mechanical Turk showed **60% of workers preferred GPT's ratings** over the original user ratings
- This suggests LLMs may offer a **less subjective** measure of user sentiment

Future
- This approach supports the development of a**utomated app recommendation system**s to aid user decision-making
- The methods have **broad applicability** to other domains, including entertainment and e-commerce review systems

ACKNOWLEDGEMENT

This work began at a hackathon hosted by the University of Delaware's DSI and AICoE. We thank Dr. Salvatore Giorgi for his feedback through the ICWSM 2025 Mentoring Program. Moath Erqsous was supported by the University of Bisha, and Fatimah Alhassan by SACM.

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Presented at AAAI ICWSM (June 23–26, 2025) Copenhagen, Denmark
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